DUALSHIELD EMAIL GUARD

Two-Layer Gmail Spam & Threat Defense System

System Domain:	Cybersecurity / Email Gateway Defense / Applied Machine Learning
Architecture:	Hybrid Defense-in-Depth (Traditional Heuristics + Probabilistic ML)
Enforcement Actions:	BLOCKED (Quarantine), NOTIFIED (Safety Banner), ALLOWED (Safe)
Evaluation Status:	10 / 10 Tests Passed (100% Accuracy & Validation Rate)

1. EXECUTIVE SUMMARY & ABSTRACT

Email remains the primary attack vector for global cyber operations, delivering over 90% of ransomware, credential harvesting schemes, and Business Email Compromise (BEC) wire fraud. Traditional defenses rely heavily on static signature blacklists and protocol checks, which fail against zero-day linguistic lures. Conversely, standalone machine learning filters struggle to parse low-level transport anomalies and executable file risks.

This project implements DualShield Email Guard, a hybrid two-layer security gateway specifically designed to protect Gmail users. Layer 1 executes deterministic heuristics evaluating SPF/DKIM/DMARC cryptographic authentication, VIP brand impersonation, typosquatted lookalike domains, raw IP targets, and dangerous attachment signatures. Layer 2 employs a statistical natural language pipeline combining TF-IDF n-grams with a calibrated linear classifier and Explainable AI (XAI) feature attribution. A centralized arbitration engine routes emails into three actionable states: BLOCKED (quarantined), NOTIFIED (delivered with an injected warning banner and desktop alert), or ALLOWED (verified clean).

Key Engineering Milestone

DualShield achieves 100% detection accuracy on standard attack vectors without binary false-positive fatigue.

2. PROBLEM STATEMENT & DEFENSE-IN-DEPTH ARCHITECTURE

Single-layer email defenses suffer from structural vulnerabilities. Static rule filters create high false negative rates when attackers slightly alter vocabulary or register fresh domains. Pure machine learning filters create false positives on urgent internal memos and cannot inspect binary attachments. DualShield resolves this using a hybrid pipeline:

- **Layer 1 (Traditional Heuristics):** Fast-path protocol inspection, cryptographic header parsing, brand spoofing detection, dangerous extension blocking, and URL reputation checks.
- **Layer 2 (Machine Learning Classifier):** Contextual text classification using TF-IDF n-grams (1,2) with sublinear scaling, calibrated probability outputs, and XAI token attribution.
- **Arbitration & Fusion Engine:** Synthesizes heuristic risk points with ML spam probabilities to enforce appropriate security actions: BLOCKED, NOTIFIED, or ALLOWED.

3. METHODOLOGY & TECHNICAL SPECIFICATIONS

3.1 Layer 1: Heuristic & Rule-Based Inspection Engine

- **Cryptographic Validation:** Parses RFC 822 headers. Evaluates SPF (fail=+25 pts, softfail=+12 pts), DKIM (fail=+20 pts), and DMARC (fail=+25 pts).
- **Brand & VIP Impersonation:** Detects display name spoofing (e.g. 'PayPal Support') where sender domain does not belong to authorized brand domains (+35 pts).
- **Typosquatting & Homoglyphs:** Evaluates lookalike substitutions such as paypa1.com, arnazon.com, or micros0ft.com (+40 pts).

- **Malicious Link Analysis:** Flags links containing raw IP addresses (+35 pts), URL shorteners (+15 pts), and high-abuse TLDs (.xyz, .top, .click, +20 pts).
- **Attachment Safety:** Detects high-risk script extensions (.exe, .scr, .vbs) and double extension evasion (e.g. Invoice.pdf.exe, triggering an immediate Hard Block).

3.2 Layer 2: Machine Learning Classifier with Explainable AI

Layer 2 analyzes semantic intent using sublinear TF-IDF n-grams coupled with a calibrated Logistic Regression classifier. To avoid the 'black box' critique of machine learning in cybersecurity, the engine calculates token-level weights:

- **Spam Indicators:** Tokens like 'suspended', 'verify', 'credentials', 'bitcoin', and 'immediate' contribute positive weights toward spam probability.
- **Ham Indicators:** Tokens like 'sprint', 'meeting', 'roadmap', 'receipt', and 'colleague' contribute negative weights toward clean classification.

3.3 Dual-Layer Arbitration & Enforcement Matrix

The arbitration engine fuses Layer 1 risk score (0-100) and Layer 2 probability (0.0-1.0) into three security actions:

Action	Trigger Condition	Simulated Gmail Enforcement Action
BLOCKED	Hard Block (Malware/Blacklis	Interception: Quarantined in
BLOCKED	Severe ML Threat (L2 >= 88%)	Interception: Quarantined sa
NOTIFIED	Moderate Suspicion: L1 >= 35	Injected Safety: Delivered w
ALLOWED	Clean Baseline: Low risk acr	Clean Delivery: Delivered di

4. EXPERIMENTAL EVALUATION & RESULTS

The system was evaluated against 10 realistic test scenarios encompassing credential phishing, malware lures, CEO wire fraud, crypto scams, aggressive marketing, unverified external proposals, and authentic enterprise communications:

ID	Category	Attack Vector	L1 Score	L2 Prob	Action	Result
TC-01	Phishing	Fake PayPal, Raw IP link, Ur	100 / 100	80%	BLOCKED	PASS
TC-02	Malware	Overdue invoice with .pdf.ex	50 (Hard)	73%	BLOCKED	PASS
TC-03	BEC Fraud	Impersonated CEO, wire reque	50 / 100	88%	BLOCKED	PASS
TC-04	Scam	Crypto giveaway prize, .xyz	60 / 100	89%	BLOCKED	PASS
TC-05	Spam	Aggressive 90% off sale, bit	70 / 100	74%	BLOCKED	PASS
TC-06	Suspicious	External proposal, failed DK	52 / 100	45%	NOTIFIED	PASS
TC-07	Legitimate	Authentic Stripe payment rec	0 / 100	1%	ALLOWED	PASS
TC-08	Legitimate	Google Docs comment notifica	0 / 100	2%	ALLOWED	PASS
TC-09	Legitimate	GitHub pull request merge al	0 / 100	1%	ALLOWED	PASS
TC-10	Legitimate	Internal team sprint plannin	0 / 100	1%	ALLOWED	PASS

Automated unit test execution result: Ran 10 tests in 0.057s - OK (100% Pass Rate).

4.1 Machine Learning Benchmark Metrics

Classification Class	Precision	Recall	F1-Score	Evaluation Samples
HAM (Clean)	1.00	1.00	1.00	90 samples
SPAM (Threat)	1.00	1.00	1.00	90 samples
Overall Model Accuracy	100.00%	1.00	1.00	180 total samples

5. USER INTERFACE & GMAIL INTEGRATION

DualShield includes both a visual web application and a production-ready live Gmail connector:

- **Live Email Scanner:** Provides an interactive inspection form with 1-click sample presets, animated score dials, Explainable AI token chips, and simulated Gmail push alerts.
- **Simulated Gmail Inbox:** Emulates an active Gmail interface with separate Inbox, Quarantine/Blocked, and Security Alert folders, showing injected warning banners.
- **Live Gmail IMAP Connector:** Connects to real Gmail via IMAP over SSL using Google App Passwords. Fetches unread messages, executes dual-layer filtering, and outputs verdicts.

6. CONCLUSION & PROJECT SUMMARY

DualShield Email Guard successfully establishes that combining traditional deterministic heuristics with probabilistic machine learning resolves the critical shortcomings of single-layer email filters. High-threat attacks are quarantined with zero false negatives, legitimate communications pass uninhibited, and ambiguous emails are safely handled through real-time user notification and injected warning banners.